

Saagar Arya

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EDUCATION

Duke University – Pratt School of Engineering

Durham, NC

B.E. Electrical & Computer Engineering / Computer Science

Aug. 2023 – Dec. 2026

GPA: 3.94 | Dean's List (Spring 2024, Fall 2024) | Distinction (Fall 2023, Fall 2025)

TECHNICAL SKILLS

Languages & Frameworks: C++, Python, C, Java, SQL, React, Bash

Infrastructure & Tools: ROS2, Kubernetes, Bazel/Starlark, Docker, Helm, GTest, Protobuf

Machine Learning: TensorRT, Ultralytics YOLO, OpenCV, PyTorch, scikit-learn, Roboflow, LabelMe

Hardware & EDA: KiCad, Eagle, Autodesk Fusion 360, SolidWorks

EXPERIENCE / PROJECTS

AeroVironment – Loitering & Munition Systems

June 2026 – August 2026

Software Engineering Intern

San Diego, CA

- Developed autonomy and task planning software in ROS2 for Switchblade aircraft
- Built and deployed a new tracker fusing vision and onboard sensor data, improving track continuity through occlusion and rapid camera motion
- Optimized the C++ pipeline to run within the 30 Hz real-time budget on flight hardware

SpaceX – Starship Communications

January 2026 – April 2026

Software Engineering Intern

Hawthorne, CA

- Migrated Starship ground software into a Kubernetes cluster, taking end-to-end ownership of infrastructure and supporting services
- Containerized applications with Docker, deployed as Kubernetes pods, and implemented cluster-level secrets handling
- Developed comprehensive unit and regression tests for the ECEF Sender application, enabling hardware-out-of-the-loop testing and replacing a costly, time-intensive hardware-in-the-loop dependency
- Built in C++ and Python using Bazel/Starlark; gained hands-on exposure to network protocols and real-time communication systems

Northrop Grumman Mission Systems – GATR Program

May 2025 – August 2025

Embedded Software Engineering Intern

Annapolis, MD

GATR (Generalized Automatic Target Recognition) parses large volumes of underwater vehicle sensor data into layered digital maps of the ocean floor

- Developed features and bug fixes in C++, Python, and XML
- Used Google Protobuf for RPC message serialization across distributed system components
- Built and maintained a GTest suite to validate processing logic and support continuous integration

Co-Founder | Blue Skull Group LLC | Utility Patent Pending

August 2023 – Present

Founded a medical device startup developing a digital pressure monitoring system for skull clamps, a neurosurgical tool used to hold the head rigid during brain surgery

- Designed and fabricated a load cell retrofit system compatible with existing skull clamp models; mapped deformation readings to calibrated force values
- Developed custom PCBs with digital potentiometers, instrumentation amplifiers, and low-noise microprocessor circuits; transmits real-time readings via Bluetooth and serial
- In active discussions with medical device companies; pursuing FDA device testing to bring the product to market

President | Duke Robotics Club

August 2023 – Present

Lead a 40-member team building an autonomous underwater vehicle (AUV) for the annual international RoboSub competition

- Oversaw integration across electrical, mechanical, and software subteams to ensure full-system cohesion
- Led full design and build of a second AUV from scratch; more compact and hydrodynamic form factor with improved maneuverability; vehicles communicate to enable coordinated task execution
- Built a hydrophone array on a custom PCB for underwater acoustic localization; developed a signal processing pipeline (Butterworth bandpass filtering, time-of-arrival analysis) feeding a Random Forest classifier that resolves pinger direction and distance
- Achieved 7th place out of 56 international teams at RoboSub 2025, the club's first-ever finals appearance

Marine Robotics & Remote Sensing Lab – Duke University

September 2023 – Present

Technical Research Intern

Beaufort, NC

- Applied YOLOv8 computer vision models to detect seals in drone imagery from Glacier Bay National Park, enabling automated large-scale wildlife monitoring
- Used human-in-the-loop active learning with confidence thresholding to iteratively improve dataset quality and detection performance
- Implemented a LiDAR system on an autonomous terrestrial rover to generate digital twins of terrain, including forested areas inaccessible to aerial surveys
- Co-author on 2 published papers; 1 additional paper in progress

Tarigma Corporation

June 2022 – September 2024

Software Engineering Intern

Columbus, OH

- Full-stack development in Java, React, SQL, Spring, and XML for G.E.M., a power grid monitoring application used to detect and diagnose issues in electrical infrastructure
- Built a fault locator tool enabling engineers to run fault calculations manually, particularly useful for correcting line geometry errors in real-world grid data
- Implemented admin and configuration features for monitoring distributed components of the US electrical grid

PATENTS & ENTREPRENEURSHIP

Etsy Flute Accessory Business

April 2020 – Present

U.S. Utility Patent #11587538 (Granted 11/2022)

- Invented and patented a 3D-printed flute rest; operate business end-to-end including design, manufacturing, customer service, financial reporting, and wholesale agreements with music retailers

Additional Patents: Skull Clamp Pressure Monitor — Provisional Patent (10/2024, 07/2025); Utility Patent Application (10/2025)

ADDITIONAL

Eagle Scout | High school hardware projects: custom-built quadcopter drone, combat robot, and hexapod robot/walker